STAT2402: Analysis of Observations

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Week 2

Revision of Linear Statistical Models

Outline

3.1 Introduction

Many situations require an assessment of which of the available variables affect a particular output.

Case Study 1: La Quinta Profitability

La Quinta Motor Inns, a moderately priced hotel chain oriented towards serving the business traveller, wants to determine which site locations are profitable. Location is one of the most important decisions for the hotel industry. A hotel chain that can select good sites more accurately and quickly than its competitors has a distinct advantage.

Consultants were hired to model their site location decision process. Data was collected over 3 years on 57 established inns belonging to La Quinta.

Modelling

A regression analysis revealed that

$$\text{Predicted profit} = 39.05 - 5.41 \textit{State population} + 5.81 \textit{Tarrif} \\ + 1.75 \textit{ColStudents} - 3.09\sqrt{\textit{MedIncome}}.$$

where

State population = state population (in thousands) per inn,
Tarrif = room rate for the inn,
ColStudents = number of college students (in thousands) within 6.5 km,
MedIncome = median income of the area.

The equation predicts that profitability will increase when *Tarrif* and number of college students **increases**, and when state population and median income **decrease**.

Case Study 2: Sales versus Promotions at Pharmex

Pharmex is a chain of drugstores, and wants to determine the effectiveness of is promotional activities. Data is collected from 50 randomly selected metropolitan locations. There are two variables:

Promote = Pharmex's promotional expenditure as a percentage of those of the leading competitor,

Sales = Pharmex's sales as a percentage of those of its leading competitor.

Objective: To quantify the relationship between sales and promotional expenses.

3.1 Introduction

Regression is the study of relationships between variables, and is a very important business tool because of its wide applicability. **Simple linear regression** involves only two variables:

X = independent or explanatory variable;
Y = dependent or response variable;

and they are related by a straight line.

The observations are $(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$.

Example 1.1

X = height of person, Y = weight

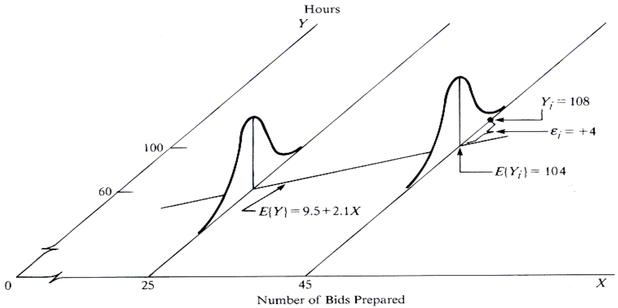
People of the same height can have different weights.

On average as height increases, weight also increases. Given the height, the weight has random variations from some mean weight for that height.

3.2 The Gaussian (Normal) linear model

A simple Gaussian linear regression model:
$Y_i = 9.5 + 2.1x_i + \epsilon_i, \quad \epsilon_i \sim N(0, \sigma^2)$

FIGURE 1.6 **Illustration of Simple Linear Regression Model (1.1).**



The Model

The general univariate statistical mode is

$$y_i = \mu_i + \epsilon_i,$$

where μ_i is the mean of y_i and ϵ_i is the random variation term.

The Model

Linear regression assumes that μ_i is a linear function of X, that is

$$\mathbb{E}(Y_i) = \mu_i = \beta_0 + \beta_1 X_i.$$

THE MODEL

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i,\ i = 1, 2, \cdots, n$$

where

Y_i =response (or dependent) variable,

X_i =explanatory (or independent) variable,

β_0 = intercept,

β_1 = slope,

ϵ_i = error or random variation

.3 Model Assumptions Observed data (X_i, Y_i), and

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, 2, \ldots, n.$$

We treat Y_i as *random variables* corresponding to observations X_i

ASSUMPTIONS

1. A linear model is appropriate, that is, $\mathrm{E}\,(Y_i) = \beta_0 + \beta_1 X_i$.
2. The error terms ϵ_i are normally distributed.
3. The error terms ϵ_i have constant variance.
4. The error terms ϵ_i are uncorrelated.

Variance

Note that

$$\mathrm{Var(Y_i) = Var(\beta_0 + \beta_1 X_i + \epsilon_i)}.$$

But $\beta_0 + \beta_1 X_i$ is a constant, so

$$\mathrm{Var(Y_i) = Var(\epsilon_i)} = \sigma^2,$$

assumed constant (that is, σ^2 does not depend on i). The assumptions then imply that

$$Y_i \overset{iid}{\sim} \mathrm{N}\left(\beta_0 + \beta_1 \mathrm{X_i}, \sigma^2\right).$$

Further,

$$\mathbb{E}\left(\epsilon_i\right) = \mathbb{E}\left[Y_i - (\beta_0 + \beta_1 X_i)\right] = 0,$$

since $\mathbb{E}(X - \mu) = 0$. This implies that

$$\epsilon_i \overset{iid}{\sim} \mathrm{N}(0, \sigma^2).$$

3.4 Detailed analysis of Pharmex data

```
pharmex <- read.table("Data/Pharmex.txt", header = T, stringsAsFactors = T)
phmx.lm <- lm(Sales ~ Promote, data = pharmex)
summary(phmx.lm)
```

```
##
## Call:
## lm(formula = Sales ~ Promote, data = pharmex)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -17.3069  -4.9544  -0.4544   5.1214  17.7423
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  25.1264    11.8826   2.115   0.0397 *
## Promote       0.7623     0.1209   6.304  8.6e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.395 on 48 degrees of freedom
## Multiple R-squared:  0.4529, Adjusted R-squared:  0.4415
## F-statistic: 39.74 on 1 and 48 DF,  p-value: 8.601e-08
```

Interpreting model output

Fitted model equation is

$$\hat{Sales} = 25.1264 + 0.7623 Promote.$$

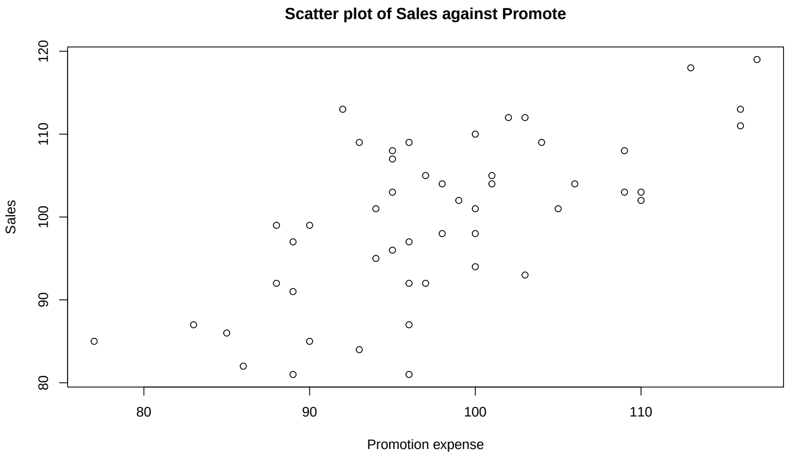
- If Pharmex spends nothing on promotion then its sales are 25% of that of its competitor.
- For every percent increase is promotion, sales increase by 0.76 percent of that of its competitor.
- If Pharmex spends the same as its competitor on promotion (that is, 100%) then its sales are 101.7% of that of its competitor.

3.5 Model diagnostics

1. The model looks quite good.The correlation coefficient is $\sqrt{0.4415} = 0.6645$, which is reasonable.
2. $r^2 = 0.4415$, so about 44% of the variation is sales is explained by expenditure on promotions.
3. BUT, we first need to verify model assumptions, that is, we need to perform model diagnostics.

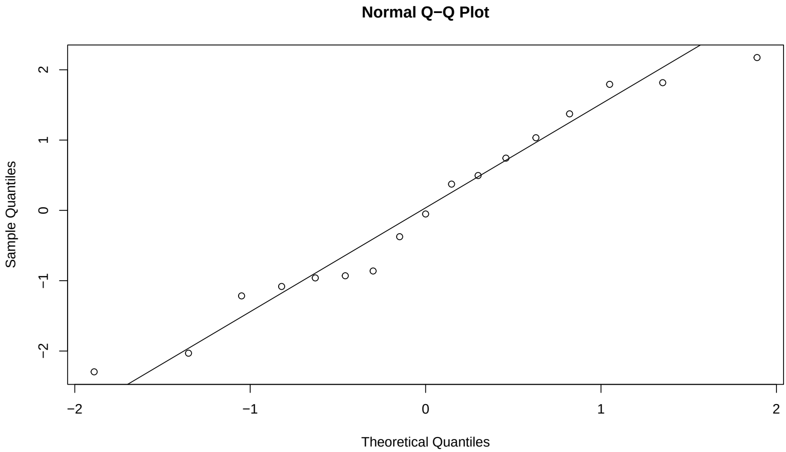
Using primitive commands

```
plot(pharmex$Sales ~ pharmex$Promote, main = "Scatter plot of Sales against Promote",
    xlab = "Promotion expense", ylab = "Sales")
```



Using ggplot

```
library(ggplot2)
phmx <- ggplot(pharmex, aes(x = Promote, y = Sales)) + geom_point()
phmx
```



Data summary

The scatterplot indicates a linear relationship is feasible.

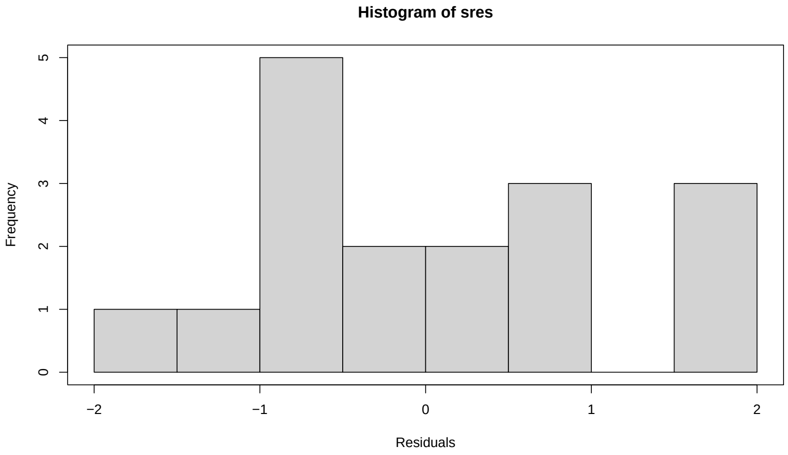
```
summary(pharmex)

##      Region         Promote           Sales
##  Min.   : 1.00   Min.   : 77.00   Min.   : 81.00
##  1st Qu.:13.25   1st Qu.: 93.00   1st Qu.: 92.25
##  Median :25.50   Median : 96.50   Median :101.00
##  Mean   :25.50   Mean   : 97.88   Mean   : 99.74
##  3rd Qu.:37.75   3rd Qu.:102.75   3rd Qu.:107.75
##  Max.   :50.00   Max.   :117.00   Max.   :119.00
```

The data range for Promotions is 77-117 with mean 97.88 and median 96.5, so promotional expenditure exceeds that of the competitor for less half the stores. Sales have a range of 81-119, with mean 99.74 and median 101, so sales exceed that of the competitor in more than half the stores.

Model diagnostics

```
resplot <- ggplot(phmx.lm, aes(x = phmx.lm$fitted.values, y = phmx.lm$residuals)) + geom_point()
resplot <- resplot + xlab("Fitted Values") + ylab("Residuals")
resplot
```

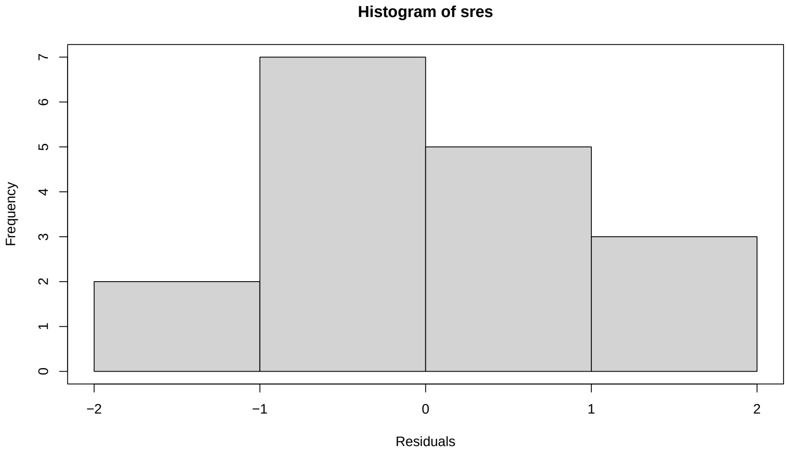


Observations

- The plot of residuals against fitted values shows no clear pattern. This indicates that a linear model is appropriate.
- However, there seems some evidence of a change in spread with fitted values—the lower values have a higher spread. That is, there the plot show "fanning in". This indicates that the homogeneous variance assumptions may not be satisfied.

Normality assumption

```r
par(mfrow = c(1, 2))
hist(phmx.lm$residuals, xlab = "Residuals")
box()
qqnorm(phmx.lm$residuals, ylab = "Residuals", xlab = "Normal Scores", main = "Normal probability plot")
qqline(phmx.lm$residuals)
```



Observations

The histogram is fairly symmetric and not unlike that expected for a normal distribution. The normal probability plot is fairly linear. Based on these we conclude there is no evidence against the normality assumption.

So overall it appears that the homogenenous variance/homoscedacity assumptions does not hold.

3.6 Parameter Estimation — Method of Least Squares

The model has three parameters, β_0, β_1 and σ^2, and these need to be estimated from the data. Denote by B_0 and B_1 respectively the estimates of β_0 and β_1.

$$\text{Fitted value } \hat{Y}_i = B_0 + B_1 X_i$$

$$\begin{aligned}\text{Residual } r_i &= Y_i - \hat{Y}_i \\ &= Y_i - (B_0 + B_1 X_i)\end{aligned}$$

We use the method of least squares to estimate B_0 and B_1 so as to minimise $\sum_{i=1}^{n} r_i^2$.

Parameter Estimates

$$\hat{B}_1 = \frac{2\mathrm{SS}_{XY}}{2\mathrm{SS}_X}$$
$$B_0 = \bar{Y} - B_1\bar{X} \Rightarrow \bar{Y} = B_0 + B_1\bar{X}.$$

Note This result can also be obtained using calculus.